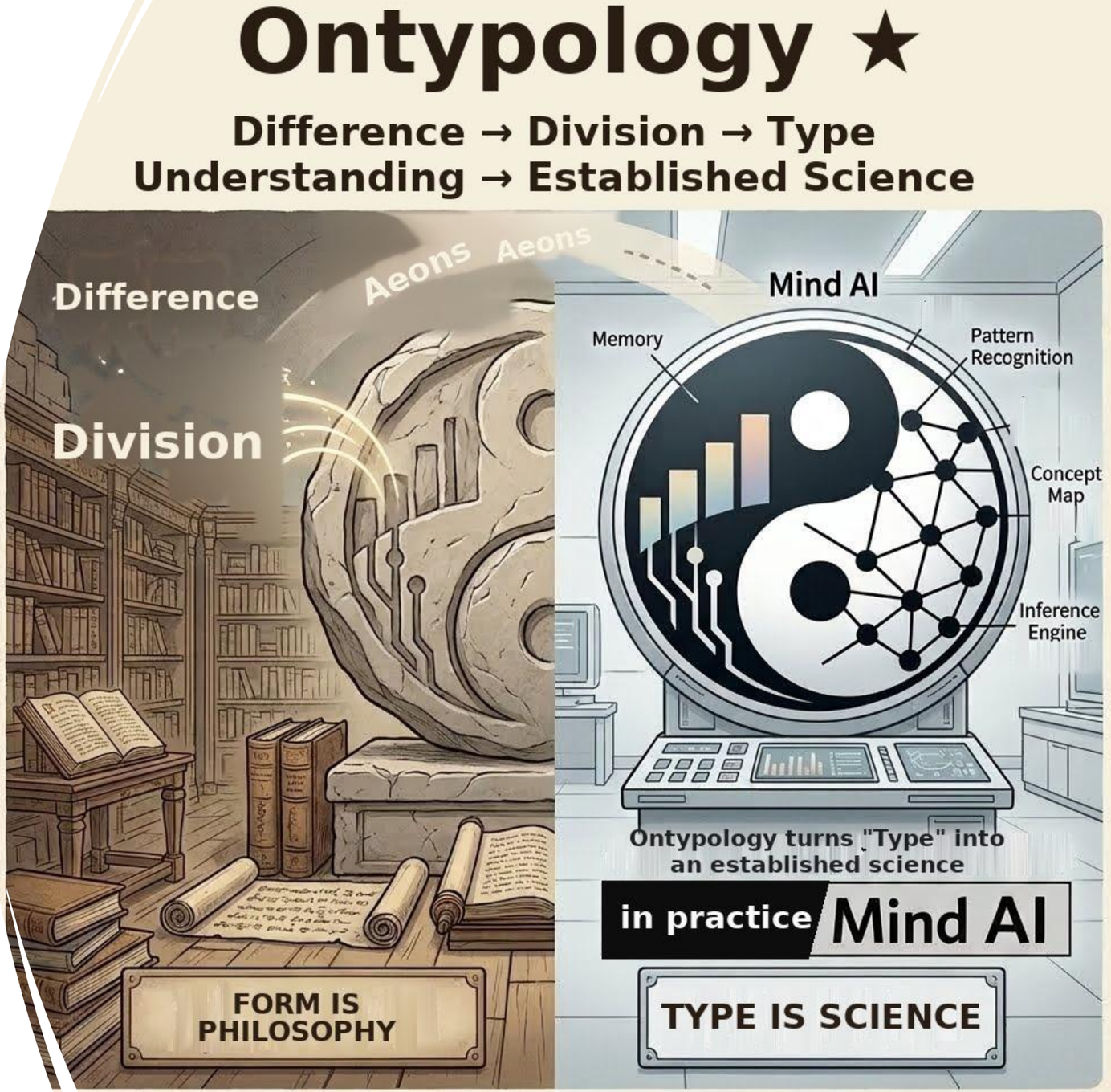
Ontypology

- From difference to division, from division to type, from type to understanding – the brain has walked this path for aeons.
- A century ago, logic was philosophy; once seen clearly, it became an established science.
- Today we explore "Type" (a brain concept) as philosophy; Ontypology makes "Type" clear, turns it into an established science, and puts it into practice as Mind AI.





AI Gene R&D Brain:

From random exploration to a reasonable R&D twin

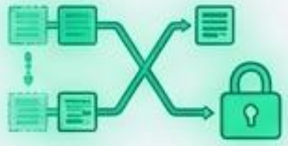
Target, guide, assess, verdict, evolution: full-path accuracy in practice

MIND AI

[[Target drift]



Symptom:
Target gene and variant site swapped midway, with no record

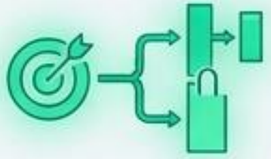


Solution:
The Target Type establishes an invariant identity id(C).

[Guide loss]



Symptom:
The sequence is right, but the PAM or the window is not covering the change.



Solution:
The Guide Type locks the ∩(T) acceptance threshold.

[Off-target miss



Symptom:
Only editing efficiency is reported, hiding off-target risk.



Solution:
The Assessment Type requires all five; none may be missing.

[No verdict]



Symptom:
The lead decision is only verbal, and next quarter nothing can be checked.



Solution:
The Verdict Type and the Evolution Type establish the evidence chain.

The common root of inaccurate R&D: the output is never written as a "Type", and the layers are not clearly written.

MIND AI

Dual-Engine Reactor

[LLM (inductive statistics)]

ID: ENG-LLM-V2



Language understanding, knowledge recall, literature extraction, drafting

R&D thinking =

Form

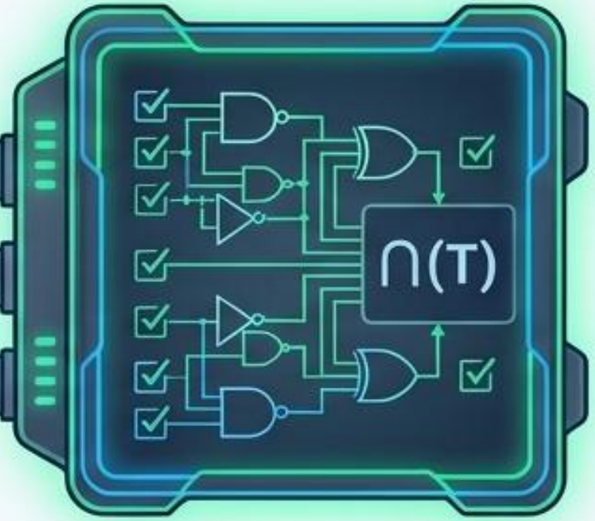
(structure, composition, evolution)

+ Number

(quantity, relation, regularity)

[Mind Model (deductive reasoning)]

ID: ENG-MM-V1



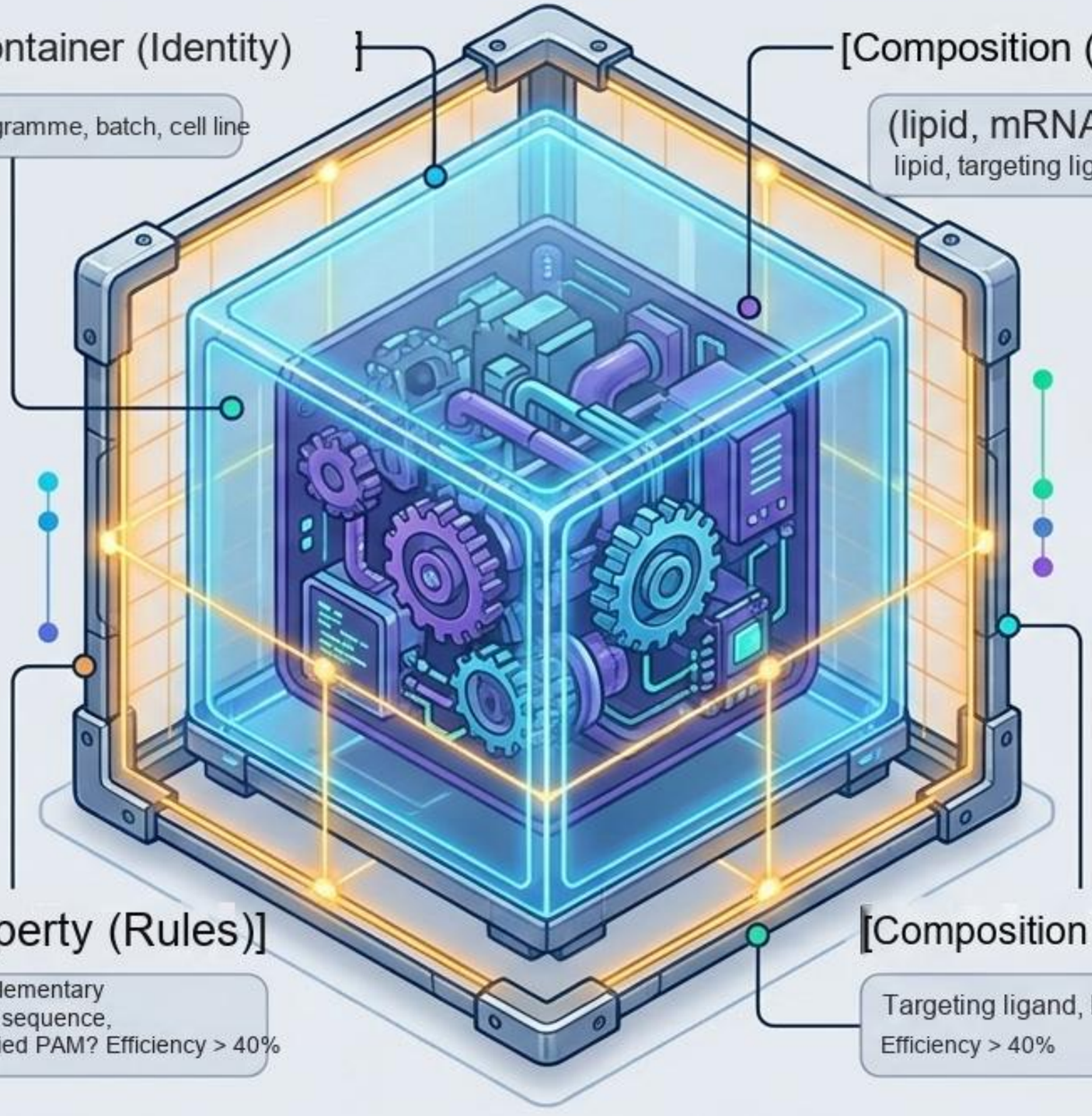
Deep reasoning, acceptance criteria ∩(T), maintain the Type evidence chain and memory

ALERT: STRATEGIC INSIGHT

A brain cannot think by induction alone. Put R&D knowledge into a frame made of Type and Number, only then can you get an R&D twin that can be questioned, rechecked, and handed on.

MIND AI

Bio-Computational Control Room Dashboard



BOM graded verification (Remove 3-Levels)

Level 1: Essential

$remove(p, M) \neq M$

Example: Cas9 nickase mRNA

Level 2: Degraded

$remove(p, M) \equiv M' \subset M$

Example: targeting ligand (targeting is lost after removal, but delivery still works)

Level 3: Non-essential

remove(p, M) and M

Example: reporter gene

Gemini Notebook

[Tactical Move Palette]

[Search set (finds it)]

ID: SEARCH-V1
1. Type-time function search

ID: SEARCH-V2
2. Design search and extraction

ID: SEARCH-V1
3. Problem solving (troubleshoot off-target anomalies)

ID: SEARCH-V4
4. R&D and exploration

[Reasoning set (thinks clearly)]

ID: AMBER-V1
5. Reasoning (phenotype inference)

ID: AMBER-62
6. Thinking methods (first principles)

ID: AMBER-74
7. Analysis (efficacy and SAR analysis)

[Design set (makes it)]

ID: EMERALD-V1
8. Design (pegRNA / vector)

ID: EMERALD-V3
9. Design patterns (screening funnel)

ID: EMERALD GREEN
10. Logic generation and application

ID: EMERALD GREEN
11. Function generation and application

[Evolution set (grows)]

ID: DEEP VIOLET-V1
12. Continuous learning (data feedback)

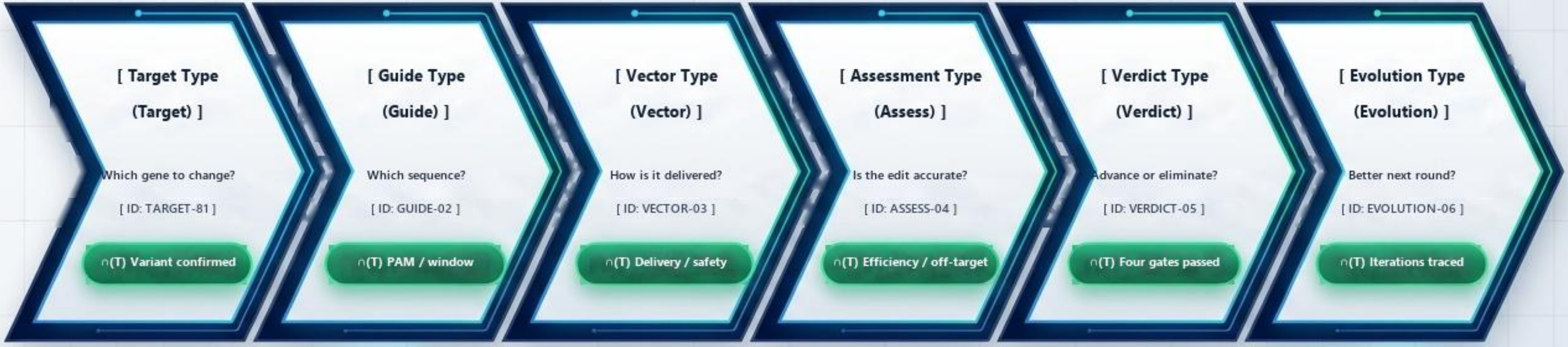
ID: DEEP VIOLET-V3
13. Refactoring and recombination

ID: DEEP VIOLET-V4
14. Self-evolution (criteria upgrade)

⚠ The fourteen moves are not a rigid lookup table but follow the R&D situation: a dynamic engine of "select, assemble, swap moves"

MIND AI

[Bio-Computational Value Chain]



[Underlying Logic]

Every Type shares the same underlying logic:

[ ] writes field rules, { } writes composition, ( ) writes identity, ∩(T) writes the intersection criterion.

Only when the whole chain runs is the R&D story clear.

MIND AI

Bio-Computational Control Room Dashboard

[Target Type]

ID: GX-2610

Target gene: HBB (beta-globin) [OMIM 603903]	✓
Variant site: HBB c.17A>T (p.Glu6Val, rs334)	✓
Indication: sickle-cell disease (SCD)	✓
Design goal: correct the point mutation by prime editing (GTG→GAG)	✓
Variable content S(C): batch cells and reagent lot numbers may be replaced	✓

[Guide Type]

Candidate: PE-HBB-01

Complementary target: 20 nt, fully complementary, 0 mismatches	✓
PAM: NGG (Cas9 nickase)	✓
Editing window: +4 to +8 covers c.17	✓
PBS/RTT: PBS 13 nt / RTT 44 nt	✓
nick site: +3	

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[Bio-Computational Value Chain: Vector & Assessment Control Room]

[Vector Type]

[Electroporation] [Chosen]

Electroporation module
nickase mRNA
pegRNA
buffer
targeting ligand

[Level 2: function-degraded]

[AAV6 donor template (Control)]

[Assessment Type - 3-part validation]

Efficiency (precise correction rate)



[Pass] measured 68% (indel 2.1%)

Off-target (GUIDE-seq)



[Pass] OT1 0.3%, OT2 0.1%

Safety (phenotype / survival)



[Pass] survival 86%, HbS down to 6%

MIND AI

Bio-Computational Control Room Dashboard

[Verdict Type]

Verdict: proceed to lead optimisation (conditional pass)

Basis: all four ∩(T) gates passed (PE-HBB-01) ✓

All nine cells complete; the Type evidence chain has no break

Condition: complete the in vivo assessment within 60 days; large-animal model scheduling

Signature	Signature	Signature
[R&D director] ●	[QA manager] ●	[Biotech safety] ●

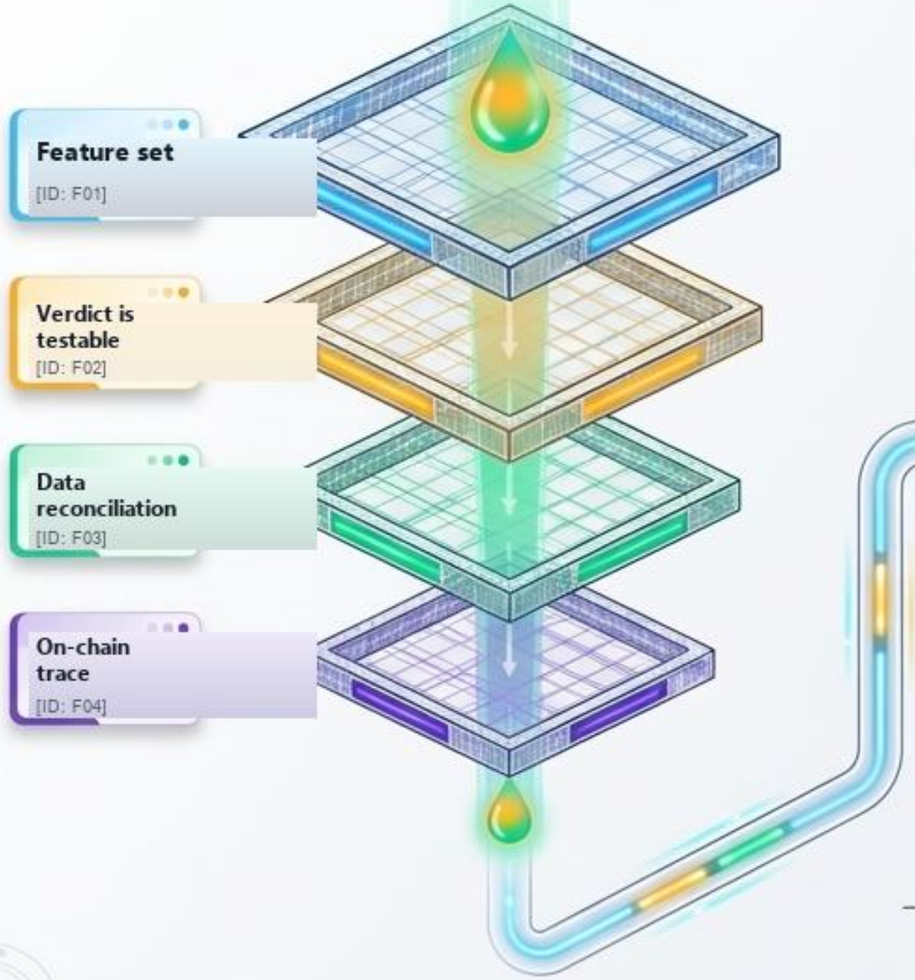
[Evolution Type (Round 2)]



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Verdict funnel ∩(T)

[Filterayer]



[Horizontal Skeleton]



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Six stations of the Type evidence chain (EvidenceChain)



Three questions per station: what were the criteria? Did it pass? Where is the evidence? Any break is returned.

Space x Time Grid

	[ ] Property layer (Rules)	{ } Composition (BOM)	( ) Container (Identity)
Design gate (AMBER)	✓ Write the features	✓ Part list	Each item tested ✓ Changing it leaves id unchanged
Evidence gate (AMBER)	✓ Full specification	✓ Full comparison	✓ Confirm the change
Decision gate (AMBER)	✓ Data archiving	✓ Risk labelling	✓ Version consistency

Only when all cells are filled does the verdict document deserve a signature. A blank cell means no verification.

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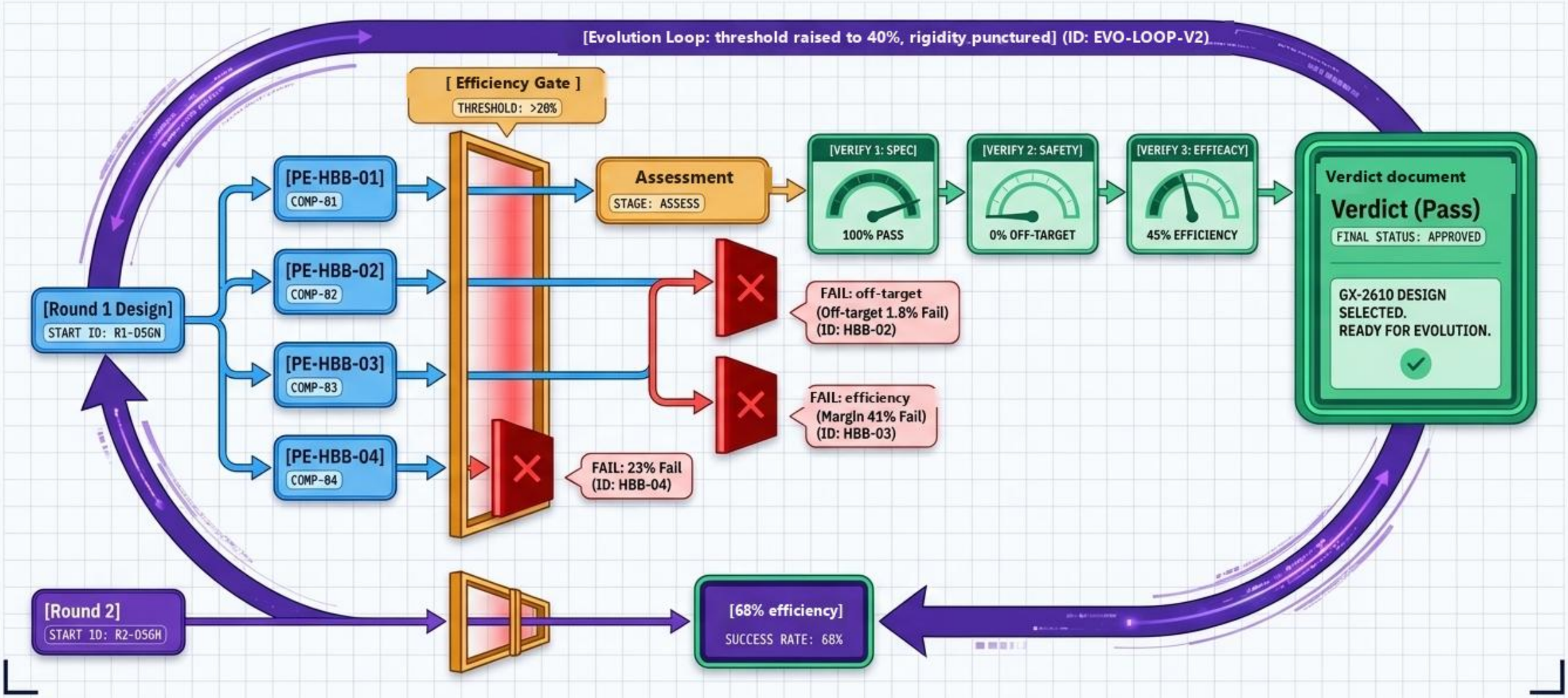
S1-S7 seven-stage relay operation card (workflow pipeline)

[Input: requirement]	S1 situation framing (four framework steps)	[Output: requirement card]
[Input: requirement card]	S2 build Types by axis (three brackets)	[Output: six-Type first draft]
[Input: Type first draft]	S3 select the expansion pattern (fourteen moves)	[Output: move selection]
[Input: move selection]	S4 generate the output (fill the fields)	[Output: output document]
[Input: measured data]	S5 verdict convergence (funnel)	[Output: convergence]
[Input: converged]	S6 cross verdict (meeting)	[Output: verdict document]
[Input: verdict document]	S7 archive and evolve (record)	[Output: evolution ledger]

Warning: the previous stage's output is the next stage's input; if any stage fails, return on the spot, and cross-stage jumping is strictly forbidden

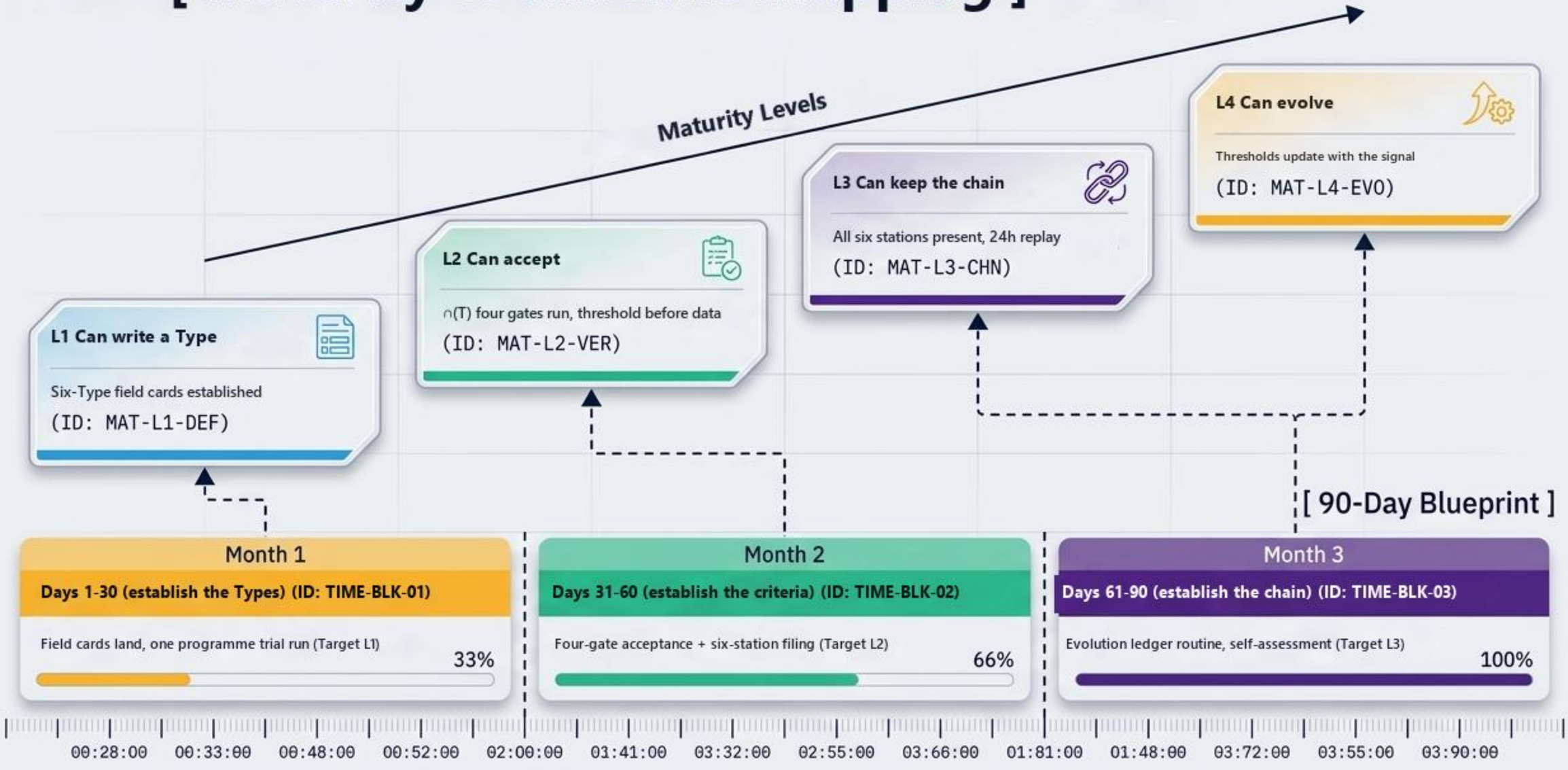
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[Full Case Panorama (GX-2610 Full Pipeline View)]



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[Maturity & Timeline Mapping]



[System Status: OPTIMAL | Data Stream: ACTIVE | Last Update: 14:02:35]

MIND AI

[Bio-Computational Control Room Dashboard]

[The Ten-Question Checklist]

- ☐ 1. Is the identity id(C) of the Target Type established? (ID: CHK-01)
- ☐ 2. Are the three brackets [ ] { } ( ) in the right places? (ID: CHK-02)
- ☐ 3. Was the threshold set before the data came out? (ID: CHK-03)
- ☐ 4. Three-piece set (efficiency, off-target, safety) has data? (ID: CHK-04)
- ☐ 5. Is the Type evidence chain unbroken, replayable anytime? (ID: CHK-05)

All ten must pass before the verdict is signed.

(ALL TEN MUST PASS TO ISSUE VERDICT. FAIL ONE QUESTION, THE COST OF A WRONG SIGNATURE IS ONE PROJECT.)

[Red Line Compliance = ∩(T)]

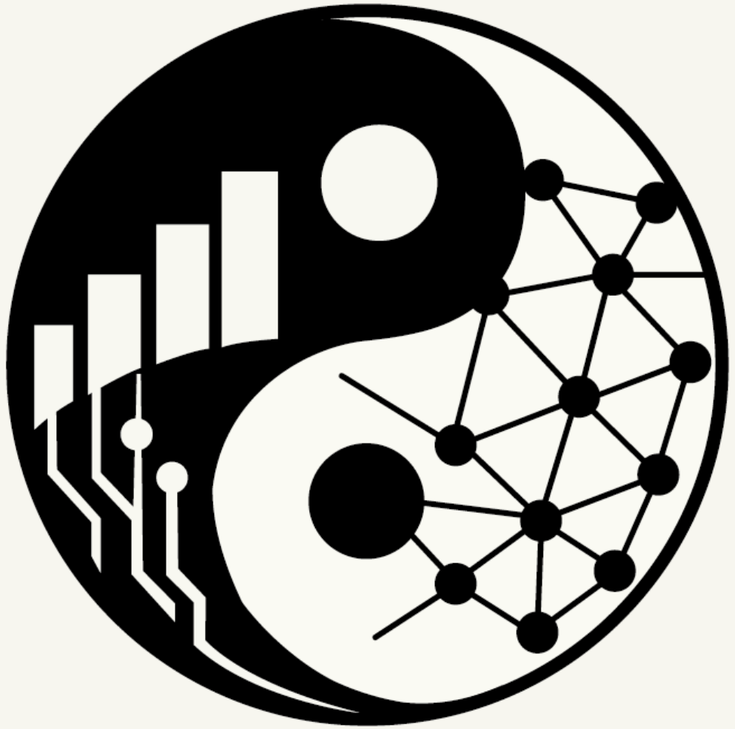
- **Germline editing: banned from clinical workflow** (ID: HAZ-01-GERM)
  (GERMLINE EDITING: CLINICAL TRIAL PROHIBITED)
- **Somatic cell therapy: strictly follow IND** (ID: HAZ-02-SOMA)
  (SOMATIC THERAPY: STRICT IND COMPLIANCE REQUIRED)
- **Long-term follow-up: condition in the evidence chain** (ID: HAZ-03-LTFU)
  (LONG-TERM FOLLOW-UP: MUST BE INCORPORATED INTO EVIDENCE CHAIN AS CONDITIONAL)
- **Informed consent: required field of the Target Type** (ID: HAZ-04-IC)
  (INFORMED CONSENT: MANDATORY FIELD, NO EXEMPTIONS)

The accuracy of gene editing includes absolute accuracy toward the red lines. The red line is the criterion: fail it and you fail the gate.

(THE ACCURACY OF GENE EDITING INCLUDES ABSOLUTE ACCURACY ON THE RED LINES. RED LINES ARE THE VERDICT; IF YOU CAN'T PASS THE RED LINES, YOU CAN'T PASS AT ALL.)

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LonGreat Vision: to create intelligent life that understands the essence of the world



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"We aim to make artificial intelligence not merely a tool, but a system of intelligent life that truly possesses the abilities to understand, reason, learn, and create."

From the unification of knowledge, to the unification of thought, and finally toward the unification of intelligence – this is the eternal path to AGI that LonGreat pursues.

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